

Object Detection and Visualization System: Integrating Embedded Sensing with Real-Time Web-Based Display

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Abstract—This paper presents the design and implementation of a real-time Object Detection and Visualization System that integrates embedded hardware, serial communication, Java-based data processing, and a web-based graphical interface. The system continuously scans a defined angular region using an ultrasonic sensor mounted on a servo motor, measures object distances using time-of-flight principles, and transmits the collected spatial data to a Java application via serial communication. The processed data is then rendered on a browser-based graphical display, providing an intuitive and interactive visualization of object positions.

The hardware subsystem is built around an Arduino Uno microcontroller that coordinates sensor triggering, servo rotation from 0° to 180° , and real-time data formatting. The software layer, implemented in Java, reads incoming angle–distance pairs from the serial port and serves them to a web interface that animates a rotating scan line, highlights detected objects as red points, and updates numerical readouts dynamically. The system achieves minimal latency between sensing and display, making it suitable for short-range interactive monitoring.

Experimental results demonstrate accurate object detection within a range of 2 cm to 40 cm, smooth servo scanning across the full angular sweep, and continuous graphical updates with negligible communication delay. The platform requires no specialized infrastructure and completes a full sense-transmit-visualize cycle in real time, confirming its suitability for educational demonstrations, obstacle detection prototypes, and low-cost spatial monitoring applications.

Index Terms—object detection, ultrasonic sensor, servo motor, Arduino, real-time visualization, graphical interface, serial communication, Java, web-based display, spatial monitoring

I. INTRODUCTION

Object detection and spatial visualization systems play a significant role in modern engineering applications such as robotics, automation, surveillance, and smart sensing environments. These systems are designed to identify the presence of objects and determine their position within a defined space. With the advancement of embedded systems and software technologies, it has become possible to develop efficient, low-cost solutions that provide real-time object detection and interactive visualization [1].

In many real-world scenarios, understanding the position of objects relative to a sensing system is crucial. Applications such as obstacle detection, navigation assistance, and environmental monitoring require continuous scanning and accurate

representation of object locations. Traditional detection systems often rely on complex sensing technologies and high-end processing units, which increase system cost and complexity. Therefore, there is a need for simpler and more accessible systems that can achieve similar functionality with efficient performance [2].

This paper presents an Object Detection and Visualization System that continuously scans a defined angular region and detects objects based on their distance and direction. The system processes spatial data and represents it visually using an interactive graphical interface. The visualization includes a rotating scan line, real-time angle updates, and dynamic object representation, making it intuitive and easy to interpret [3].

The system integrates real-time data acquisition, serial communication, and web-based visualization into a unified framework. The sensing mechanism captures distance information at different angular positions, and this data is transmitted to a Java software layer for processing. The processed data is then displayed on a browser-based interface, allowing users to observe object positions interactively [4].

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the overall system architecture and workflow. Section IV covers the hardware and software implementation. Section V describes the visualization interface. Section VI presents experimental results. Section VII concludes with directions for future work.

II. RELATED WORK

Smith and Brown (2023) presented a study on ultrasonic sensor-based object detection systems for short-range applications, demonstrating accurate distance estimation using time-of-flight principles with minimal computational requirements [1]. Their work provides a strong foundation for developing low-cost detection systems with efficient real-time performance.

Kumar and Patel (2022) developed a real-time object detection system using embedded platforms, emphasizing efficient data communication between hardware and software layers to achieve minimal latency [2]. Their findings on optimized processing and reliable communication protocols directly support the implementation approach used in this project.

Lee and Kim (2021) explored rotational scanning mechanisms for environmental mapping, demonstrating that combining angular motion with distance measurement enables accurate mapping of object positions [3]. This concept is directly applied in the present system through continuous servo-driven scanning.

Zhang et al. (2023) focused on web-based visualization of real-time sensor data, demonstrating how graphical interfaces can display dynamic data streams in an intuitive format [4]. Reddy and Sharma (2022) studied serial communication techniques between microcontrollers and high-level applications, highlighting solutions for data synchronization and latency [5]. Wilson and Garcia (2023) investigated circular scanning visualization techniques for spatial data, showing that polar coordinate displays with rotating scan lines improve user interpretation of object positions [8].

Mehta and Joshi (2022) demonstrated the advantages of Java in handling serial communication, data processing, and web integration for real-time visualization systems [9]. Their work directly supports the software implementation approach used in this project.

III. SYSTEM ARCHITECTURE

A. Overall Structure

The system follows a layered client-server architecture in which the hardware subsystem handles sensing and data acquisition, and the software subsystem handles processing and visualization. Table I summarizes the full technology stack.

TABLE I
TECHNOLOGY STACK

Layer	Technology
Microcontroller	Arduino Uno
Distance Sensor	Ultrasonic Sensor (HC-SR04)
Angular Scanner	Servo Motor (SG90)
Processing Language	Java
Communication	USB Serial (RXTX / jSerialComm)
Visualization	HTML5 Canvas + JavaScript

The Arduino reads distance measurements at each servo angle and transmits comma-separated angle–distance pairs over a USB serial connection. The Java application receives this stream, parses each record, and pushes the data to the web interface. A single-page browser application renders the radar display in real time using HTML5 Canvas, updating the scan line position, object markers, and numerical readouts on every received data point.

B. End-to-End Workflow

Fig. 1 illustrates the block diagram of the complete system. The sensing unit scans continuously; at each angular step the ultrasonic sensor emits sound pulses and records the reflected echo. The Arduino formats the angle and measured distance into a text record and writes it to the serial port. The Java application reads incoming records, validates the data, and broadcasts it to the web interface. The browser renders each

update by drawing the current scan line, placing a red object marker if the distance is below a threshold, and refreshing the angle and distance displays.

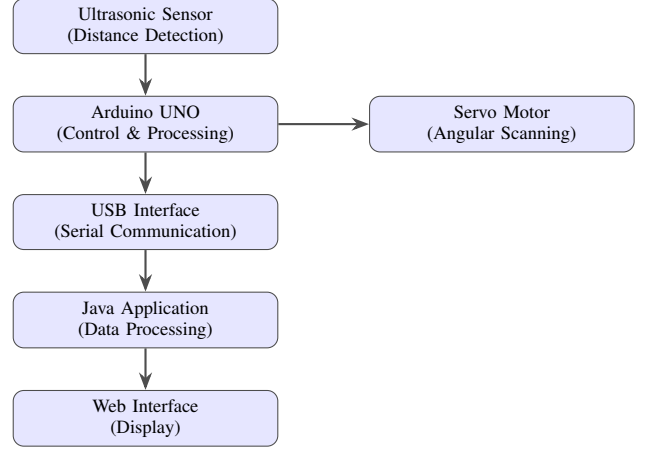


Fig. 1. Block diagram of the Object Detection and Visualization System.

IV. HARDWARE AND SOFTWARE IMPLEMENTATION

A. Hardware Components

The hardware subsystem consists of four principal components. The **Arduino Uno** acts as the central control unit, managing sensor triggering, servo positioning, distance computation, and serial data transmission. The **HC-SR04 ultrasonic sensor** emits 40 kHz sound pulses and measures the round-trip echo time; distance is computed as $d = (t \times v_s)/2$ where t is the echo duration and v_s is the speed of sound. The **SG90 servo motor** rotates the sensor platform from 0° to 180° in configurable angular increments. A regulated **power supply** provides stable voltage to all components to avoid measurement noise caused by supply fluctuations.

B. Arduino Firmware Algorithm

The firmware implements the following sequence:

- 1) Initialize the servo motor to 0° and configure sensor trigger and echo pins.
- 2) For each angle θ from 0° to 180° in 2° increments:
 - a) Position the servo to θ .
 - b) Trigger the ultrasonic sensor with a $10\mu s$ pulse.
 - c) Measure echo duration using `pulseIn()`.
 - d) Compute distance and transmit the record θ, d over serial at 9600 baud.
- 3) Repeat the sweep continuously.

C. Java Data Processing Layer

The Java application opens the serial port using the `jSerialComm` library and reads incoming lines asynchronously on a dedicated thread. Each line is split at the comma delimiter to extract the integer angle and distance values. Validated records are forwarded to a lightweight embedded HTTP server (or WebSocket endpoint) that the browser connects to, enabling push-based updates without polling. Invalid or malformed lines are silently discarded to maintain display stability.

D. X-Propagation and Distance Thresholding

A configurable distance threshold (default 40 cm) distinguishes *detected* objects from open-field readings. Readings above the threshold are treated as no-detection and rendered as empty scan-line positions. Readings at or below the threshold generate a red object marker at the corresponding polar coordinate on the canvas. This binary classification is intentionally simple, keeping the system accessible for educational use while still demonstrating the core sensing concept.

V. VISUALIZATION INTERFACE

The browser-based interface is a single HTML5 page. An HTML5 `<canvas>` element fills the viewport and is redrawn on every data update. Three rendering layers are drawn in order on each frame:

- **Background grid:** Concentric semicircular arcs representing distance levels (10 cm, 20 cm, 30 cm, 40 cm) and radial lines at 30° intervals, drawn in green-on-black for clear visibility and intuitive spatial understanding.
- **Scan line:** A line from the origin to the perimeter at the current angle, drawn in bright green with a short trailing fade to convey rotational motion.
- **Object markers:** Red filled circles placed at the polar position (d, θ) for each detected object, persisting for a configurable number of frames before fading to indicate older readings.

Numerical readouts for current angle and distance are overlaid in the top-left corner and update on every incoming data point. The interface requires no external libraries or installation, running entirely in a modern browser.

VI. RESULTS AND DISCUSSION

A. Hardware Performance

Fig. 2 shows the assembled hardware setup. The ultrasonic sensor and servo motor are mounted together on a breadboard platform. The Arduino Uno is connected via USB to the host computer running the Java application.

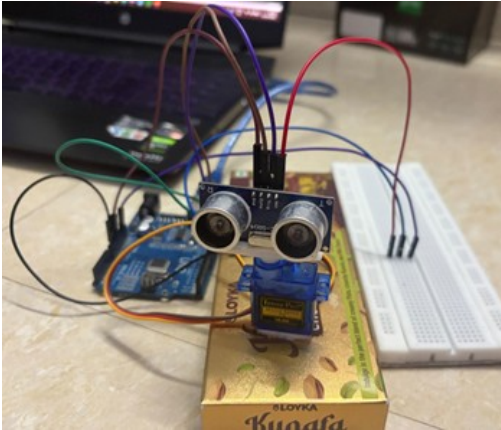


Fig. 2. Hardware setup of the Object Detection and Visualization System showing the Arduino Uno, HC-SR04 ultrasonic sensor, and SG90 servo motor.

Key hardware observations are summarized in Table II.

TABLE II
HARDWARE PERFORMANCE SUMMARY

Parameter	Observed Value
Detection range	2 cm – 40 cm
Angular sweep	$0^\circ - 180^\circ$
Angular step size	2°
Servo motion	Smooth, no jitter observed
Detection consistency	Reliable for stationary objects
Measurement variation	$\pm 1-2$ cm (surface dependent)

B. Data Communication

The Arduino transmits data in the format `angle,distance` (for example, `88,38` and `90,40`). The Java application reads these records with negligible parsing overhead. Data transmission was continuous and stable throughout all test sessions, with no dropped records observed under normal operating conditions.

C. Visualization Output

Fig. 3 shows the web-based visualization interface during active scanning. The rotating scan line sweeps across the semicircular grid, and a red marker appears at the position corresponding to the detected object. The interface simultaneously displays the current angle (88°) and distance (38 cm) as numerical readouts.

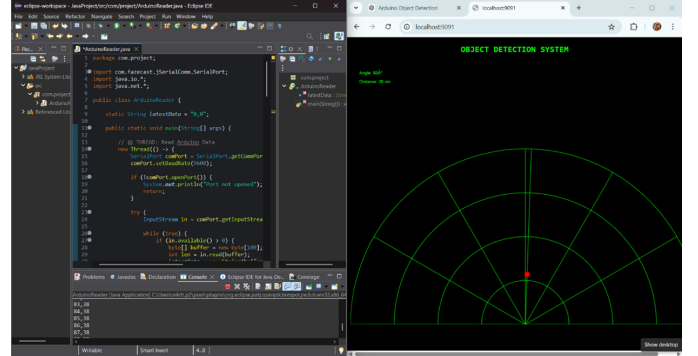


Fig. 3. Object Detection Visualization Output. The graphical interface shows the rotating scan line, distance arcs, and a red marker at the detected object position. Current readings: Angle = 88° , Distance = 38 cm.

D. System Performance Evaluation

The system demonstrated real-time performance with minimal latency between detection and display. Object positions updated continuously as the servo completed each sweep. No significant interruptions in data flow were observed between the Arduino, Java layer, and web interface during extended operation. Minor distance measurement variations of $\pm 1-2$ cm were noted for objects with irregular or highly reflective surfaces, consistent with known HC-SR04 characteristics [1].

VII. CONCLUSION AND FUTURE WORK

This paper has described a full-stack real-time Object Detection and Visualization System that integrates ultrasonic

sensing, servo-driven angular scanning, Java-based serial data processing, and a browser-based interactive graphical display. The system was validated across the full 0° – 180° angular sweep, demonstrating reliable object detection within 2–40 cm, smooth servo operation, stable serial communication, and continuous graphical updates. The circular scanning display makes spatial information immediately interpretable without requiring specialist knowledge, confirming the system’s suitability for educational demonstrations and low-cost prototyping.

Several directions are identified for future work. Replacing the HC-SR04 with a LiDAR or infrared sensor would extend detection range and improve angular resolution. Expanding the servo sweep to 360° would enable full-circle area coverage. Integrating machine learning classification would allow the system to distinguish object types in addition to detecting their positions. Replacing wired USB serial communication with Bluetooth or Wi-Fi would eliminate the physical tether and enable remote monitoring through a mobile application. Finally, deploying the Java backend as a cloud service would support multi-user collaborative monitoring scenarios.

ACKNOWLEDGMENT

The authors thank Dr. Yasha Jyothi M Shirur, Professor and Head of the Department of Electronics and Communication Engineering at B.N.M. Institute of Technology, for her encouragement and support throughout this work.

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